

We have discussed *how motion is described* in term of *velocity* and *acceleration*. Now we deal with the question of *why objects move as they do*: What makes an object *at rest begin to move*? What causes a body to *accelerate* or *decelerate*? what is involved when an object *moves in a circle*? we can answer in each case that *a force is required*. In this chapter, we will investigate the connection between *force and motion*, which is the subject called *dynamics*.

Dynamics studies the rule on the interaction force between bodies and examines the causes of motion.



Isaac Newton

Isaac Newton , jazz,the royal society member, is an England **physicists** and **mathematicians**, **astronomers**, natural philosophers and alchemy.He is the founder of classical physics. He has made great discoveries in mechanics, optics, heat, astronomy and mathematics. His representative work "mathematical principles of natural philosophy" is a classic work of mechanics. Newton is a great scientist who has made great contributions to the foundation of modern natural science

Chapter 4 Newtonian Mechanics

- 1. Newton's Law (three laws of motion)**
- 2. Some Particular Forces**
- 3. Applying Newton's Laws**

4.1 Newton's first law — inertial law

Newton's first law: any object must maintain rest or moving in average linear motion, until force acting on the object changing its motion.

while $\vec{F} = 0$, $\vec{v} = \text{constant vector}$

Supplementary explanation:

- 1) The concept of **Inertia**: The mass is an **intrinsic** characteristic of a body, it characterizes the inertia of the body.
- 2) The force is the **only reason** which **makes the states of body change**. It indicates the **concept of force**: An interaction —can cause an acceleration of a body.

4.1 Newton's first law — inertial law

3) Inertial reference frames (IRF) : An inertial frame is one in which Newton's laws hold.

Does Newton's first law hold in every reference frame? **No**

For example, if your reference frame is fixed in a car, and the car accelerates, an object such as a ball resting on the dashboard may begin to move toward you (it stayed at rest as long as the car's velocity remained constant). The ball accelerated toward you but neither you nor anything else exerted a force on it in that direction. In such an accelerating reference frame, Newton's first law **does not** hold.

4.2 Newton's second law

3.2.1 momentum

$$\vec{p} = m\vec{v}$$

4.4.2 Newton's second law : object has momentum with total external force, the changing rate of momentum of the object equals to the total outside force.

$$\vec{F} = \frac{d\vec{p}}{dt} = \frac{d(m\vec{v})}{dt} = \frac{dm}{dt}\vec{v} + m\frac{d\vec{v}}{dt}$$

So in **classical mechanics**, it can be expressed as:

$$\boxed{\vec{F}_{\text{net}} = m\vec{a}} \quad \text{-----Newton's second law}$$

The *acceleration* of an object is *directly proportional* to the *net force* acting on it and is *inversely proportional* on its *mass*. the direction of the acceleration is in the direction of the net force acting on the object.

Unit : newton (N) $1\text{N} = 1\text{kg} \cdot \text{m/s}^2$

4.2. Newton's second law

Supplementary explanation:

1. It is a law of force's **instantaneous interaction** . Force is the causes of motion.
2. It is only suitable for the **inertial frame**, and **particles** or **particle-like bodies**.
3. $\vec{F}_{\text{net}}$ is the vector sum of **all net external forces**.
4. The **component forms** of N-II Law:

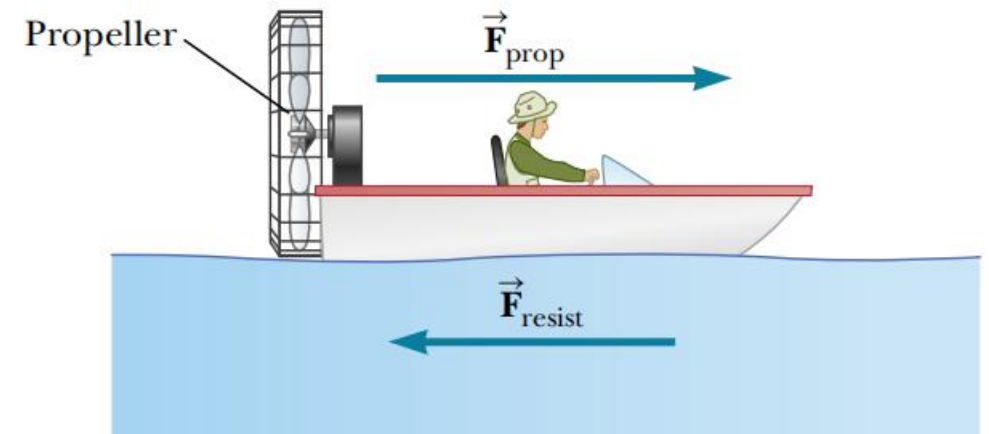
$$F_{\text{net},x} = m \sum a_{k,x}, \quad F_{\text{net},y} = m \sum a_{k,y}, \quad \text{and} \quad F_{\text{net},z} = m \sum a_{k,z}$$

— The acceleration component along a given axis is caused **only** by the sum of the force components **along that same axis**, and not by force components along any other axes.

4.2. Newton's second law

Example:

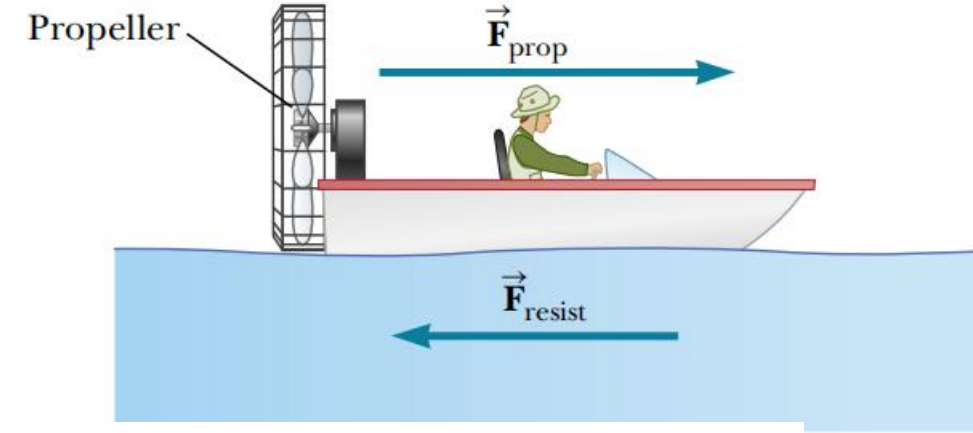
An airboat with mass $3.50 \times 10^2 \text{ kg}$, including the passenger, has an engine that produces a net horizontal force of $7.70 \times 10^2 \text{ N}$, after accounting for forces of resistance. (a) Find the acceleration of the airboat. (b) Starting from rest, how long does it take the airboat to reach a speed of 12.0 m/s ? (c) After reaching that speed, the pilot turns off the engine and drifts to a stop over a distance of 50.0 m . Find the resistance force, assuming it's constant.



4.2. Newton's second law

Example:

An airboat with mass $3.50 \times 10^2 \text{ kg}$, including the passenger, has an engine that produces a net horizontal force of $7.70 \times 10^2 \text{ N}$, after accounting for forces of resistance. (a) Find the acceleration of the airboat.



SOLUTION

(a) Find the acceleration of the airboat.

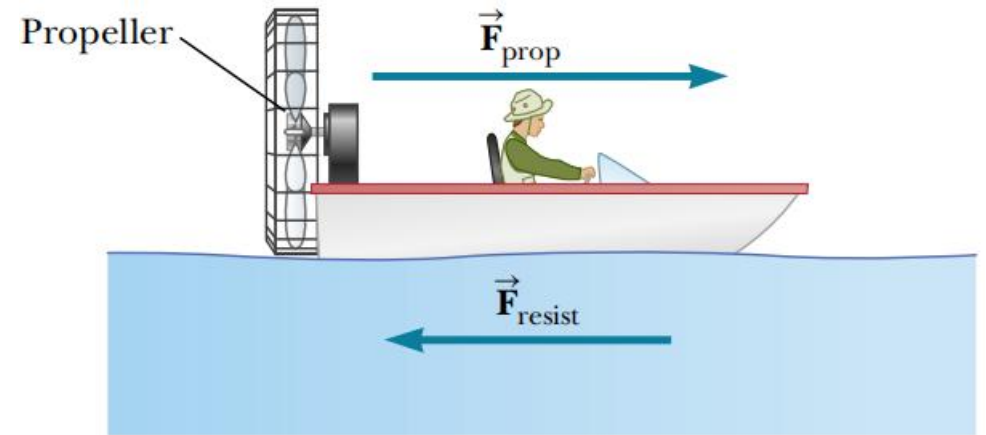
Apply Newton's second law and solve for the acceleration:

$$ma = F_{\text{net}} \rightarrow a = \frac{F_{\text{net}}}{m} = \frac{7.70 \times 10^2 \text{ N}}{3.50 \times 10^2 \text{ kg}} = 2.20 \text{ m/s}^2$$

4.2. Newton's second law

Example:

An airboat with mass $3.50 \times 10^2 \text{ kg}$, including the passenger, has an engine that produces a net horizontal force of $7.70 \times 10^2 \text{ N}$, after accounting for forces of resistance. (b) Starting from rest, how long does it take the airboat to reach a speed of 12.0 m/s ?



SOLUTION

(b) Find the time necessary to reach a speed of 12.0 m/s .

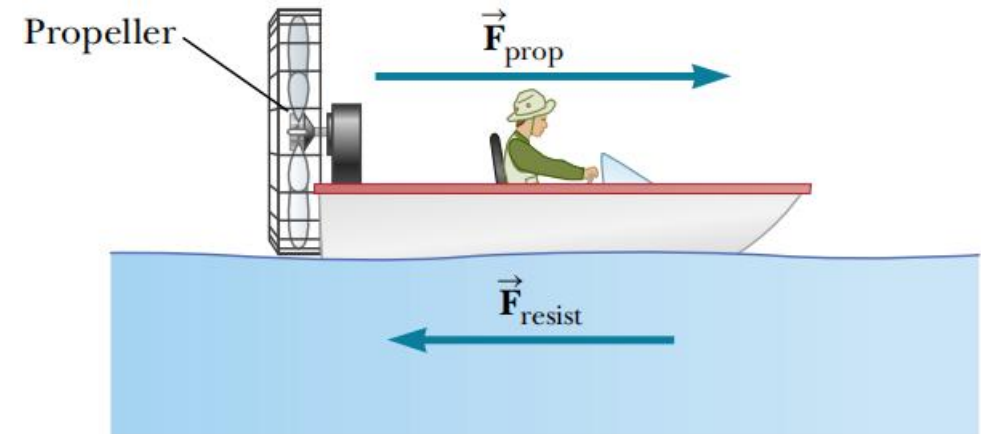
Apply the kinematics velocity equation:

$$v = at + v_0 = (2.20 \text{ m/s}^2)t = 12.0 \text{ m/s} \rightarrow t = 5.45 \text{ s}$$

4.2. Newton's second law

Example:

An airboat with mass $3.50 \times 10^2 \text{ kg}$, including the passenger, has an engine that produces a net horizontal force of $7.70 \times 10^2 \text{ N}$, after accounting for forces of resistance. (c) After reaching that speed, the pilot turns off the engine and drifts to a stop over a distance of 50.0 m. Find the resistance force, assuming it's constant.



SOLUTION

(c) Find the resistance force after the engine is turned off.

Using kinematics, find the net acceleration due to resistance forces:

Substitute the acceleration into Newton's second law, finding the resistance force:

$$v^2 - v_0^2 = 2a \Delta x$$

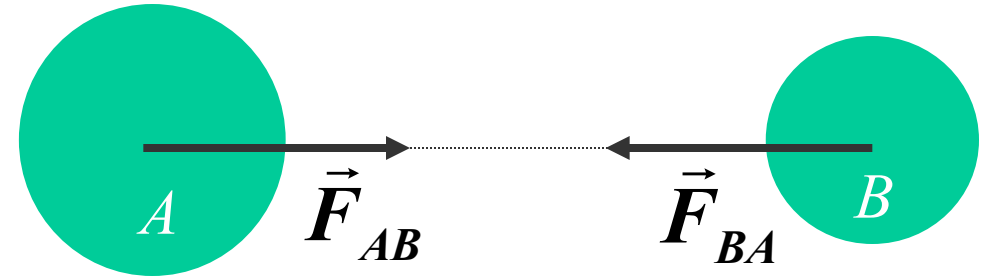
$$0 - (12.0 \text{ m/s})^2 = 2a(50.0 \text{ m}) \rightarrow a = -1.44 \text{ m/s}^2$$

$$F_{\text{resist}} = ma = (3.50 \times 10^2 \text{ kg})(-1.44 \text{ m/s}^2) = -504 \text{ N}$$

4.3. Newton's third law

Whenever one object exerts a force on a second object, the second exerts an **equal** and **opposite** force on the first. (to every action there is an equal and opposite reaction):

$$\vec{F}_{AB} = -\vec{F}_{BA}$$



Supplementary explanation:

- 1) Same type of force;
- 2) the “action” force and the “reaction” force are acting on different bodies, do not cancel each other.



The third-law force pair act on **different bodies**.

see Fig.

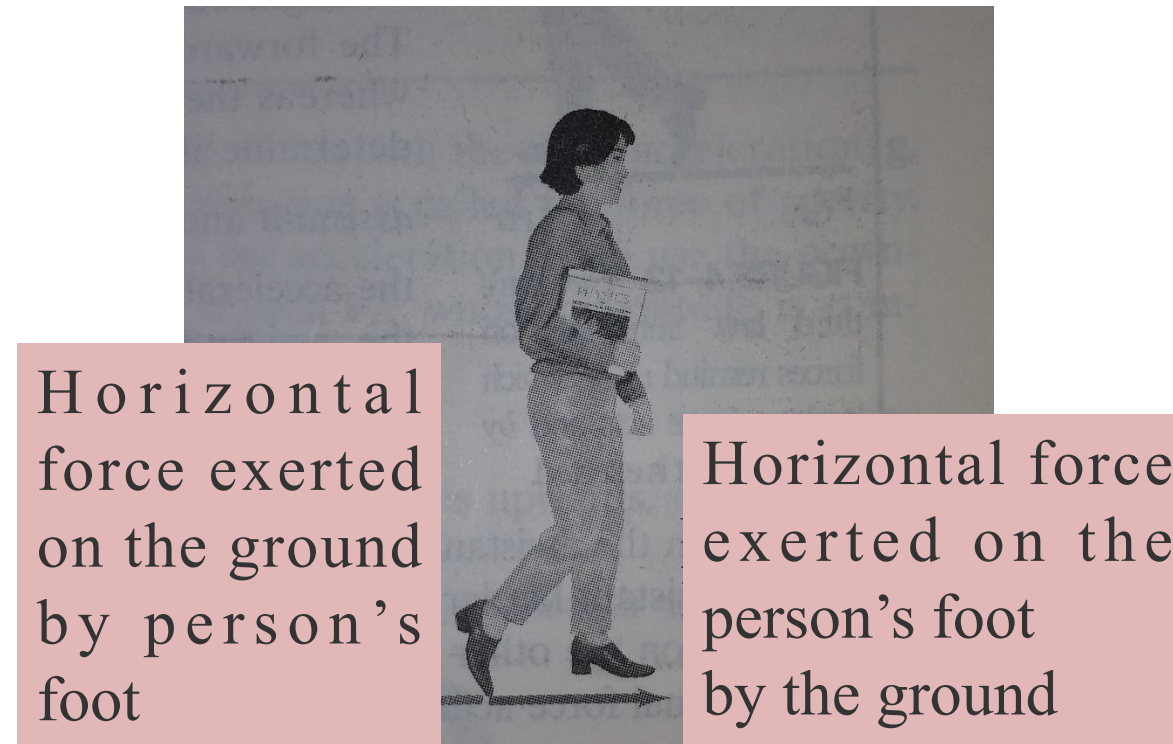


FIGURE We can walk forward because, when one foot pushes **backward** against the ground, the ground pushes **forward** on the foot. The two forces shown *act on different objects*.

4.4. The gravitational force

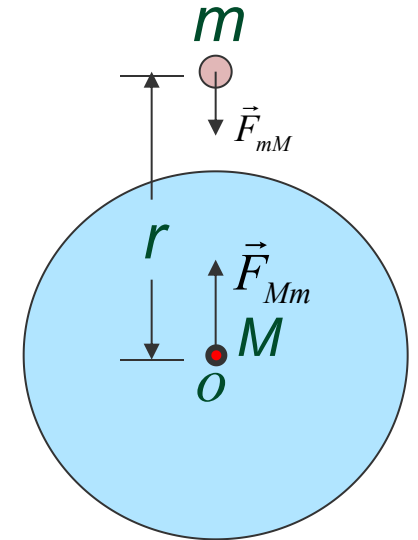
3.4.1. The gravitational force

The gravitational force is the **mutual** force of attraction between any two objects in the Universe. Although the gravitational force can be very strong between very large objects, it's the weakest of the fundamental forces.

Newton's law of universal gravitation states **that** every particle in the Universe attracts every other particle with a force that is **directly proportional to the product of the masses** of the particles and **inversely proportional to the square of the distance** between them.

Newton's form of the law of gravitation :

$$\vec{F}_{Mm} = -G \frac{Mm}{r^2} \vec{e}_r$$



where $\vec{F}_{Mm}$ is the vector force on particle M (of mass M) exerted by particle m (of mass m), which is distance r away; $\vec{e}_r$ is a unit vector, and directs away from its origin; G is the gravitational constant (**universal**).

$$G = 6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2 / \text{kg}^2$$

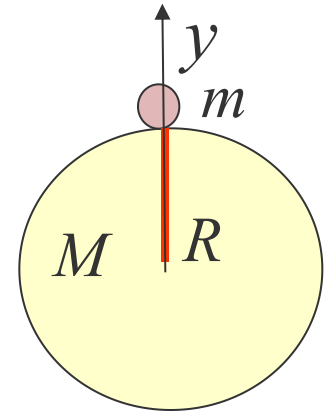
4.4. The gravitational force

3.4.2. Weight

For a body on the ground, of mass m , $\vec{F} = -G \frac{M_E m}{R^2} \hat{r} = -mg \vec{j}$

$$g = G \frac{M_E}{R^2} = 6.67 \times 10^{-11} \text{ N m/kg}^2 \frac{5.98 \times 10^{24} \text{ kg}}{(6.37 \times 10^6 \text{ m})^2} = 9.8 \text{ m/s}^2$$

where $\vec{F}_g = m\vec{g}$ is called **weight** ——— a special case of gravitation .



Unlike mass, weight is **not** an inherent property of an object because it can take different values, depending on the value of g in a given location. If an object has a mass of 70.0 kg, for example, then its weight at a location where $g = 9.80 \text{ m/s}^2$ is $mg = 686 \text{ N}$. In a high-altitude balloon, where g might be 9.76 m/s^2 , the object's weight would be 683 N. The value of g also varies slightly due to the density of matter in a given locality. In this text, unless stated otherwise, the value of g will be understood to be 9.80 m/s^2 , its value near the surface of the Earth.

4.5. Applying Newton's Laws

■ PROBLEM-SOLVING STRATEGY

Newton's Second Law

Problems involving Newton's second law can be very complex. The following protocol breaks the solution process down into smaller, intermediate goals:

1. **Read** the problem carefully at least once.
2. **Draw** a picture of the system, identify the object of primary interest, and indicate forces with arrows.
3. **Label** each force in the picture in a way that will bring to mind what physical quantity the label stands for (e.g., T for tension).
4. **Draw** a free-body diagram of the object of interest, based on the labeled picture. If additional objects are involved, draw separate free-body diagrams for them. Choose convenient coordinates for each object.
5. **Apply Newton's second law.** The x - and y -components of Newton's second law should be taken from the vector equation and written individually. This usually results in two equations and two unknowns.
6. **Solve** for the desired unknown quantity, and substitute the numbers.

Note: Try to choose one axis in the direction of the acceleration.

It may simplify the algebra.

Example:

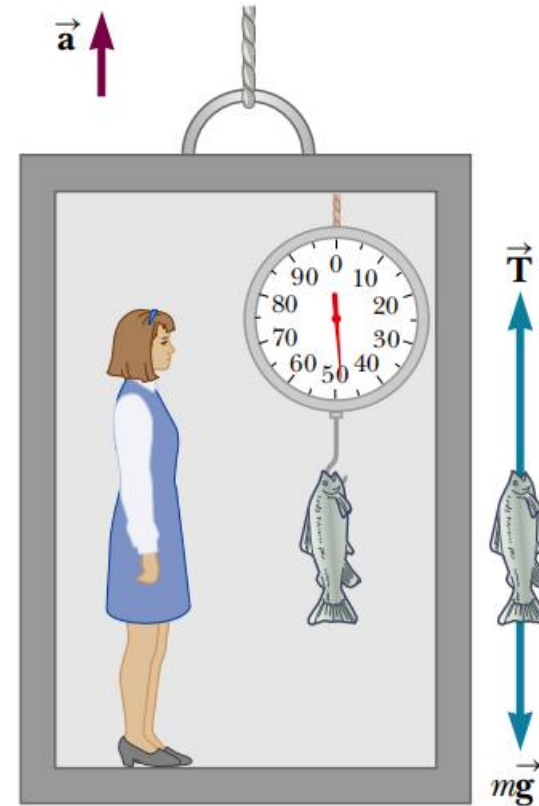
A woman weighs a fish with a spring scale attached to the ceiling of an elevator, as shown in Figures a and b. While the elevator is at rest, she measures a weight of 40.0 N.

(a) What weight does the scale read if the elevator accelerates upward at 2.00 m/s^2 ?

(b) What does the scale read if the elevator accelerates downward at 2.00 m/s^2 , as in Figure b?

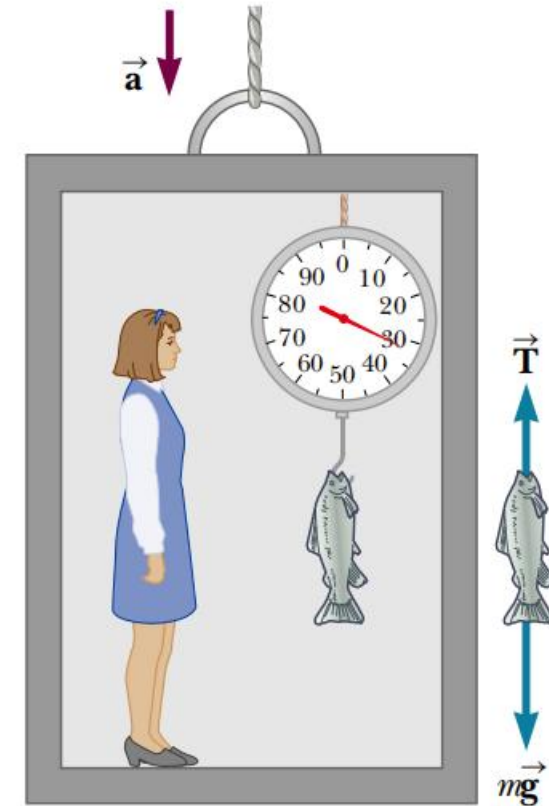
(c) If the elevator cable breaks, what does the scale read?

When the elevator accelerates upward, the spring scale reads a value greater than the weight of the fish.



a

When the elevator accelerates downward, the spring scale reads a value less than the weight of the fish.



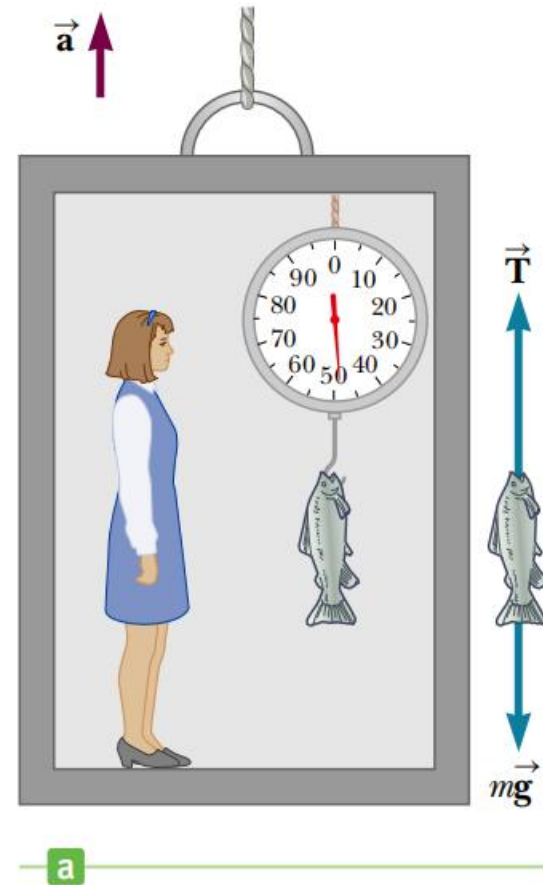
b

Example:

A woman weighs a fish with a spring scale attached to the ceiling of an elevator, as shown in Figures a and b. While the elevator is at rest, she measures a weight of 40.0 N.

(a) What weight does the scale read if the elevator accelerates upward at 2.00 m/s^2 ?

When the elevator accelerates upward, the spring scale reads a value greater than the weight of the fish.



SOLUTION

(a) Find the scale reading as the elevator accelerates upward, as in Figure a.

Apply Newton's second law to the fish, taking upward as the positive direction:

Solve for T :

Find the mass of the fish from its weight of 40.0 N:

Compute the value of T , substituting $a = +2.00 \text{ m/s}^2$:

$$ma = \sum F = T - mg$$

$$T = ma + mg = m(a + g)$$

$$m = \frac{w}{g} = \frac{40.0 \text{ N}}{9.80 \text{ m/s}^2} = 4.08 \text{ kg}$$

$$\begin{aligned} T &= m(a + g) = (4.08 \text{ kg})(2.00 \text{ m/s}^2 + 9.80 \text{ m/s}^2) \\ &= 48.1 \text{ N} \end{aligned}$$

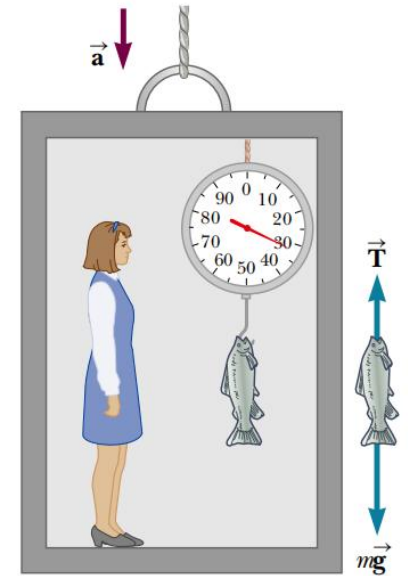
Example:

A woman weighs a fish with a spring scale attached to the ceiling of an elevator, as shown in Figures a and b. While the elevator is at rest, she measures a weight of 40.0 N.

(b) What does the scale read if the elevator accelerates downward at 2.00 m/s^2 , as in Figure b?

(c) If the elevator cable breaks, what does the scale read?

When the elevator accelerates downward, the spring scale reads a value less than the weight of the fish.



b

SOLUTION

(b) Find the scale reading as the elevator accelerates downward, as in Figure 4.19b.

The analysis is the same, the only change being the acceleration, which is now negative: $a = -2.00 \text{ m/s}^2$.

$$\begin{aligned} T &= m(a + g) = (4.08 \text{ kg})(-2.00 \text{ m/s}^2 + 9.80 \text{ m/s}^2) \\ &= 31.8 \text{ N} \end{aligned}$$

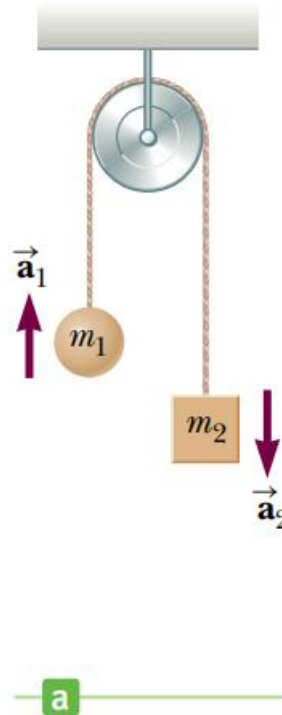
(c) Find the scale reading after the elevator cable breaks.

Now $a = -9.80 \text{ m/s}^2$, the acceleration due to gravity:

$$\begin{aligned} T &= m(a + g) = (4.08 \text{ kg})(-9.80 \text{ m/s}^2 + 9.80 \text{ m/s}^2) \\ &= 0 \text{ N} \end{aligned}$$

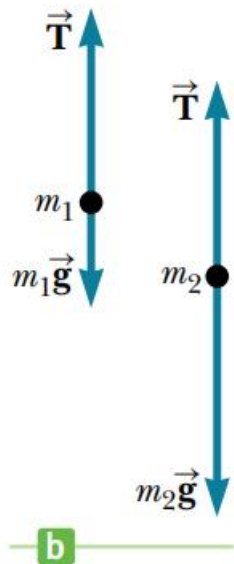
Example: Atwood's Machine

Two objects of mass m_1 and m_2 , with $m_2 > m_1$, are connected by a light, inextensible cord and hung over a frictionless pulley, as in Active Figure a. Both cord and pulley have negligible mass. Find the magnitude of the acceleration of the system and the tension in the cord.



Example: Atwood's Machine

Two objects of mass m_1 and m_2 , with $m_2 > m_1$, are connected by a light, inextensible cord and hung over a frictionless pulley, as in Active Figure a. Both cord and pulley have negligible mass. Find the magnitude of the acceleration of the system and the tension in the cord.



SOLUTION

Apply the second law to each of the two objects individually:

Substitute $a_2 = -a_1$ into Equation (2) and multiply both sides by -1 :

Add Equations (1) and (3), and solve for a_1 :

Substitute this result into Equation (1) to find T :

$$(1) \quad m_1 a_1 = T - m_1 g \qquad (2) \quad m_2 a_2 = T - m_2 g$$

$$(3) \quad m_2 a_1 = -T + m_2 g$$

$$(m_1 + m_2) a_1 = m_2 g - m_1 g$$

$$a_1 = \left(\frac{m_2 - m_1}{m_1 + m_2} \right) g$$

$$T = \left(\frac{2m_1 m_2}{m_1 + m_2} \right) g$$